Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / bending
Case	rca_20f0a28316c346d4977713d54a978394
Date	2026-11-01 18:12:03
Status	AI proposal, pending expert review

1. Problem

Defect	Severity	Confidence	Detail
crack	critical	95%	9.0 mm long

2. Root cause

Cause: Material Hardness Variation

Reason: The detected critical crack at the bend radius is directly linked to hardness variation in SPCC steel, which increases susceptibility to cracking during forming operations as described in [02cf0928-6739-4f7e-ae1b-f6740283f00f].

Effect: Cracking at bend radius

Confidence: 89%

Supporting evidence
- **02cf0928-6739-4f7e-ae1b-f6740283f00f**: Material Inconsistency / Hardness Variation Root Cause Analysis Defect Types: crack, deformation, dimensional variation Material: SPCC steel, cold-rolled sheet metal Root Cause: Hardness variation in incoming material lot causes unpredictable springback behavior and susceptibility to cracking under forming operations. Effect: Cracking at bend radius, dimensional inaccuracy due to variable springba

3. Solution

Corrective actions (fix this occurrence)

1. Verify material certificates against specification for yield strength, tensile strength, and hardness
2. Conduct incoming material inspection for hardness testing

Preventive actions (stop it recurring)

1. Implement stricter incoming quality control protocols for raw material batches
2. Audit material suppliers for consistency in cold-rolling processes

Precautions

- Monitor for batch-to-batch variation in material properties

4. Other causes considered

Rank	Cause	Confidence	Reason
2	Work Hardening Inconsistency	86%	The evidence [02cf0928-6739-4f7e-ae1b-f6740283f00f] notes that work hardening inconsistency in SPCC steel can lead to surface defects and cracking during forming.

5. Traceability

AI model: gemini:gemini-3.1-flash-lite. Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.